



ClinVar

cancer-gene missense
GRCh38 SNVs, dedup

36,075 variants

Sequence

Δ mass, Δ hydro, Δ charge
BLOSUM62, WT/MT identity
44 features

Structure


AlphaFold via UniProt
fpocket, DSSP, pLDDT
8 features

Evolution


UCSC phyloP100way
phastCons100way bigWigs
2 features

Feature matrix

$30,620 \times 54$
rows $\times$ features
1,040 genes

Gene-grouped split

70% train
15% validate, 15% test
5 seeds

ML classifier

Random Forest
MLP, XGBoost
F1, PR, ROC, ECE